

Tomato Late Blight: Identification and Management (Sample Extension Guide)

Late blight of tomato is caused by the oomycete pathogen *Phytophthora infestans*. It is one of the most destructive diseases of tomato and potato crops worldwide, capable of destroying entire fields within one to two weeks under favorable conditions.

Symptoms typically begin as small, water-soaked, pale green to dark olive spots on leaves, often starting at the leaf edges or tips. Under humid conditions, a white, fuzzy fungal growth may appear on the underside of affected leaves. Lesions rapidly enlarge and turn brown to black, and infected leaves may curl and die within a few days.

Stems can develop dark brown to black lesions that girdle the stem, causing wilting of the portion above the lesion. Fruit infection appears as firm, brown, greasy-looking blotches that can cover large portions of the fruit surface.

The disease spreads rapidly under cool, wet, and humid weather, particularly when temperatures range from 60 to 78 degrees Fahrenheit with extended periods of leaf wetness. Spores are dispersed by wind and rain splash and can travel considerable distances between fields.

Management recommendations include: planting resistant or tolerant tomato varieties where available; ensuring adequate plant spacing and pruning to improve airflow and reduce humidity within the canopy; avoiding overhead irrigation, especially in the evening, to reduce leaf wetness duration; removing and destroying infected plant debris promptly, as the pathogen can survive on plant residue; and rotating crops to avoid planting tomatoes or potatoes in the same location in consecutive seasons.

Preventive fungicide applications, timed according to local disease forecasting systems and following product label instructions, can help protect healthy plants before infection occurs. Always follow local agricultural extension guidance and product labels for specific fungicide recommendations, application rates, and pre-harvest intervals.

Farmers are strongly encouraged to consult their local agricultural extension office for region-specific management advice, as disease pressure and effective control strategies vary significantly with local climate and cropping practices.